

RAJENDRA P N

LEAD AZURE DEVOPS ENGINEER | AI | K8S

rajendran.scm@gmail.com

+1 (469) 926-4201

Dallas, TX

linkedin.com/in/rajendranelakurthi

rajendranelakurthi.github.io

Lead Azure DevOps Engineer and senior Kubernetes platform engineer with 10+ years of cloud engineering and production delivery experience. Engineers end-to-end **AKS** architecture, security, administration, observability, scaling, and upgrades using **Terraform, Azure DevOps, Helm, and Python**. Deploys GPU workloads, Milvus/Qdrant vector databases, and FastAPI microservices. Partners with engineering teams on platform standards, technical evaluations, incident troubleshooting, and operational readiness.

CERTIFICATIONS



Microsoft Certified: Azure DevOps Engineer Expert



Microsoft Certified: Azure Fundamentals



GitLab Certified Associate



MuleSoft Certified Developer - Level 1

KUBERNETES PLATFORM & RELIABILITY HIGHLIGHTS

- End-to-End AKS Engineering** Connects cluster architecture, infrastructure automation, application onboarding, security, observability, and day-to-day administration into maintainable platform services.
- Platform Lifecycle & Reliability** Plans Kubernetes upgrades around API compatibility, disruption budgets, capacity, validation, and production recovery readiness.
- AI Workload Enablement** Deploys isolated GPU capacity and persistent vector database services to support AI applications and FastAPI-based microservices.

CORE TECHNOLOGIES

AKS ARCHITECTURE & GKE KNOWLEDGE

AKS: cluster design, node pools, availability zones, private networking, Azure CNI, ingress, DNS, RBAC, workload identity, quotas; GKE knowledge: architecture, node pools, release channels, maintenance and upgrades

LIFECYCLE, SCALING & OPTIMIZATION

AKS upgrades, API compatibility, maintenance windows, surge capacity, cordon/drain, PodDisruptionBudgets, node-image updates, HPA, cluster autoscaler, requests/limits

MICROSERVICES & DELIVERY

Python, FastAPI, REST APIs, Docker, Azure DevOps YAML pipelines, Azure Container Registry, Helm, Argo CD, health probes, controlled rollouts

KUBERNETES SECURITY & OPERATIONS

Image vulnerability scanning, hardened containers, least privilege, NetworkPolicies, Key Vault integration, TLS certificates, secret rotation, incident troubleshooting

AI & VECTOR DATABASE PLATFORMS

GPU node pools, NVIDIA device plugin, GPU resource requests, taints/tolerations, node affinity, Milvus, Qdrant, Helm, StatefulSets, persistent storage, backup/restore

OBSERVABILITY & INFRASTRUCTURE AUTOMATION

Azure Monitor, Log Analytics, Prometheus, Grafana, alerts, dashboards, Terraform, Bicep, Azure CLI, Bash, PowerShell, Linux

PROFESSIONAL EXPERIENCE

Lead Azure DevOps Engineer | AT&T Services Inc.

08/2023 - Present

Skills Used: Azure, AKS, Kubernetes, Terraform, Bicep, Azure DevOps, Helm, Python, FastAPI, GPU Nodes, Milvus, Qdrant, Prometheus

- Provided senior production escalation support for scheduling, image-pull, crash-loop, storage, networking, and resource-pressure incidents; documented root causes, recovery steps, and corrective actions.
- Owned end-to-end AKS platform architecture and administration, aligning cluster capabilities, application onboarding, engineering standards, and operational runbooks with platform roadmap priorities.
- Evaluated GPU, vector-search, and deployment capabilities through proofs of concept, documenting capacity, security, supportability, and cost tradeoffs for stakeholders.
- Secured workloads with image vulnerability scanning, hardened containers, network policies, Key Vault secrets, and TLS lifecycle management; coordinated remediation and certificate/connectivity troubleshooting.
- Tuned requests/limits, HPA, cluster autoscaler settings, and node-pool sizing to balance stability, performance, and cost; reviewed capacity requirements with application teams.
- Provisioned AKS, system/application node pools, networking, registry, and identity integrations with Terraform and Bicep; administered namespaces, RBAC, workload identity, and quotas.
- Deployed Milvus and Qdrant through Helm; administered persistent storage, resource sizing, access controls, backup procedures, and restore validation for vector-search services.
- Planned AKS control-plane and node-pool upgrades with API-deprecation checks, add-on compatibility testing, maintenance windows, surge capacity, PodDisruptionBudgets, node-image updates, and staged validation.
- Deployed GPU node pools for AI applications, validating NVIDIA device-plugin readiness, GPU resource requests, node affinity, and taints/tolerations for workload isolation.
- Delivered FastAPI and REST microservices through Azure DevOps, Azure Container Registry, and Helm, configuring health probes, environment settings, ingress routing, and controlled rollouts.
- Built Azure Monitor, Log Analytics, Prometheus, and Grafana dashboards and alerts for node health, pod restarts, deployment status, application latency, and capacity.

Senior DevOps Engineer | GlobalLogic

03/2023 - 08/2023

Skills Used: Kubernetes, Docker, Helm, Git, Linux, CI/CD

- Partnered with development and QA teams to standardize Kubernetes deployment practices, troubleshoot runtime issues, and improve engineering handoffs.
- Engineered Kubernetes workload deployment patterns with Helm and version-controlled manifests, aligning application configuration with environment and operational standards.
- Integrated container image validation and security checks into delivery pipelines and partnered with developers on remediation and deployment readiness.
- Administered namespaces, services, ConfigMaps, secrets, and resource settings and diagnosed pod scheduling, container startup, and service-connectivity failures.
- Maintained Linux and Windows build agents and reusable delivery workflows for repeatable image builds, automated tests, artifact publication, and release approvals.
- Used workload health checks and staged rollouts to verify application changes before promotion and documented recovery steps for failed releases.

Senior DevOps Engineer II | Akamai Technologies

12/2021 - 03/2023

Skills Used: Azure, AKS, Terraform, Helm, Argo CD, GPU Nodes, Python, Azure Monitor

- Managed Kubernetes workloads, node-pool capacity, deployment configuration, and secure application connectivity across engineering environments.
- Supported GPU-enabled AKS application deployments, validating workload placement, resource allocation, and operational telemetry with application teams.
- Provisioned Azure and AKS infrastructure with Terraform and ARM templates and deployed containerized services using Helm and Argo CD.
- Planned AKS upgrade readiness by reviewing version compatibility, deployment manifests, health probes, resource capacity, and maintenance requirements.
- Correlated Azure Monitor, Log Analytics, and pipeline telemetry to isolate pod, networking, infrastructure, and deployment failures.
- Built Python and PowerShell automation for environment validation, deployment checks, and recurring cluster administration tasks.
- Maintained release runbooks, versioned artifacts, and post-deployment checks to support consistent operational handoffs.

Senior DevOps Engineer / Senior SRE | Turvo India Pvt Ltd

04/2020 - 12/2021

Skills Used: Azure, AKS, Terraform, Python, FastAPI, Jenkins, JFrog, Prometheus, Grafana

- Built reusable Terraform modules for Azure networking and AKS, standardizing infrastructure provisioning and application deployment patterns.
- Used Prometheus and Grafana telemetry to troubleshoot application and platform issues and support reliable production releases.
- Automated Linux and deployment operations with Ansible and Python and documented platform support and recovery procedures.
- Packaged and deployed microservices through Kubernetes and Helm, aligning application configuration, environment secrets, health checks, and release dependencies.
- Administered CloudBees Jenkins, JFrog Artifactory, and Xray and integrated SonarQube checks and secure artifact promotion into delivery workflows.
- Developed Python FastAPI integrations for Jenkins, Jira, and JFrog to automate onboarding, release coordination, and operational workflows.

Senior DevOps Engineer | HCL Technologies

02/2019 - 04/2020

Client: Mercedes-Benz Research & Development India

Skills Used: Azure, Jenkins, ARM templates, Ansible, Helm, Key Vault

- Built Jenkins and Maven pipelines for REST microservices across testing, security validation, UAT, and production environments.
- Investigated application and deployment failures with ELK, Azure Monitor, and Log Analytics and coordinated verified service recovery with engineering teams.
- Packaged Kubernetes services with Helm and managed environment configuration for repeatable deployment and operational support.
- Integrated Key Vault and secure database connectivity and supported certificate, credential, and application-connectivity troubleshooting.
- Automated Azure virtual machines, networking, and Linux configuration with ARM templates, Ansible, and PowerShell.

DevOps Engineer | Mphasis Ltd

08/2016 - 12/2018

Client: HPE

Skills Used: Azure, Terraform, Ansible, Python, PowerShell, Linux

- Created repeatable pipelines for API, database, and UI components and maintained Git branching practices and build-agent configuration.
- Automated Azure infrastructure and application provisioning with Terraform, ARM templates, and Ansible.
- Troubleshot Linux, build, configuration, and release failures with development and operations teams and documented recovery steps.
- Developed Python and PowerShell scripts for deployment preparation, administration, and environment diagnostics.

EDUCATION

B.Tech (2009-2013) in Information Technology

Prakasam Engineering College, Jawaharlal Nehru Technological University, Kakinada